

Too Easy Fault Injection Attacks on Learning with Rounding (LWR)

Francesco Berti
EnICS, Selecsys Labs
Bar-Ilan University
Ramat-Gan, Israel
francesco.berti@biu.ac.il

Sasha Petri
TU-Darmstadt
Darmstadt, Germany
sasha.petri@outlook.de

Itamar Levi
EnICS, Selecsys Labs
Bar-Ilan University
Ramat-Gan, Israel
itamar.levi@biu.ac.il

Abstract—We present an extend-and-prune fault Injection attack on serial implementations of Learning With Rounding that drop the least-significant bits. By iteratively isolating and recovering progressively larger key portions via faults, the attack recovers the secret key.

Index Terms—cryptography, hardware security, fault injection, learning with error, LWR

I. INTRODUCTION

One of the most popular countermeasures against side-channel attacks (SCA) [1] is *masking* [2]–[5], which splits a sensitive value into several shares and performs the computation on these shares. Unfortunately, such countermeasures can be very expensive for block-ciphers, as the overhead is typically quadratic in the number of shares (or the security level). Alternatively, there are cryptographic primitives which can be used for (e.g.,) encryption that can be masked more easily. For example Learning Parity with Noise (LPN) [6], and its physical counterparts, Learning Parity with Physical Noise (LPPN) [7]–[11], or Learning with Rounding (LWR) [7]. In these primitives, the output is a “noisy” version of the inner product $\mathbf{k} \cdot \mathbf{x}$ of the key $\mathbf{k}$ with a random vector $\mathbf{x}$. The output is made noisy, either by the addition of a random bit, that equals 1 with probability $\epsilon < 1/2$ (LPN), or by making the computation incorrect with probability ϵ by running the circuit computing $\mathbf{k} \cdot \mathbf{x}$ at too low voltages or too high frequency (LPPN [8]), or by giving a rounded version of the output (LWR). For typical security levels LWR constructions require computing dot-product on *thousands* of elements, *each* represented by 16/32-bits. Note that in the context of this work which focuses on serial implementations, the motivation is clear as parallelism in such constructions would be quite limited in most software and hardware platforms.

An adversary having physical access to a device performing a cryptographic computation, can obtain useful information by injecting faults [12]. Fault injection (FI) attacks can be highly precise using localized injections. On advanced process nodes, attackers can even target individual registers or gates, for example via Laser L-FI [13] or electromagnetic EM-FI [14], [15]. Berti and Standaert [16] have proved that it is possible to attack a serial implementation of LPN by setting some bits

to 0 (whereas faults that flip are useless). Their idea is to isolate a key nibble, then, use a maximum-likelihood strategy to recover it. Then, they apply an extend-and-prune strategy, that is, they isolate a bigger key piece containing the key nibble recovered, and they use a maximum likelihood strategy to recover it, exploiting the key nibble already recovered. This attack remains effective even under imprecise fault locations (although performance degrades).

Our contributions: We consider the case where the key $\mathbf{k} \in \mathbb{Z}_{2^q}^n$ and the rounding outputs is the p most significant bits of $\mathbf{k} \cdot \mathbf{x}$. We prove that it is possible to apply the same strategy as Berti et al. [16] against a serial implementation of LWR¹. We show that an attack is possible *even when only one bit can be faulted per computation*.

II. PRELIMINARIES

We denote by $x \stackrel{\$}{\leftarrow} \mathcal{X}$ sampling x uniformly from $\mathcal{X}$. Let $\mathbf{x}$ be a vector in $\mathbb{Z}_{2^q}^n$, and we denote its components by $\mathbf{x} = (x_1, \dots, x_n)$. Let $x_{i,j}$ be the j^{th} bit of the i^{th} component of $\mathbf{x}$. Note that $x_i = \sum_{j=0}^{q-1} 2^j x_{i,j}$. We denote with $(\mathbf{k}, \mathbf{x})$, often simplified with $\mathbf{k} \cdot \mathbf{x}$, the inner product.

Learning with Rounding (LWR): Given $q', p' \in \mathbb{N}, q > p$, the *rounding function* $\lfloor \cdot \rfloor : \mathbb{Z}_{q'} \rightarrow \mathbb{Z}_{p'}$ is defined as $\lfloor x \rfloor := x \cdot p' / q'$. When $q' = 2^q$ and $p' = 2^p$, the rounding function has a very easy description: $\lfloor x \rfloor := x_{q-1} \dots x_{q-p}$, where x is the string $x_{q-1} \dots x_0$. Note that we use x_i to denote the i^{th} coordinate of the vector $\mathbf{x}$, while $x_{i,j}$ denotes the j^{th} most significant bit of x_i (to avoid confusion we use $x_{i,j}$ to denote the j^{th} most significant bit of x).

Given $q, p, n \in \mathbb{N}$ with $q > p$, given a random $\mathbf{k} \stackrel{\$}{\leftarrow} \mathbb{Z}_{2^q}^n$, the LWR *distribution* $\text{LWR}_{n,p,q,\mathbf{k}}$ is defined as $(\mathbf{x}, \lfloor \mathbf{k} \cdot \mathbf{x} \rfloor)$ with $\mathbf{x} \stackrel{\$}{\leftarrow} \mathbb{Z}_{2^q}^n$. The *decision problem* is to distinguish the $\text{LWR}_{n,p,q,\mathbf{k}}$ from a random distribution and the *search problem* is to find $\mathbf{k}$, given $\text{LWR}_{n,p,q,\mathbf{k}}$ [7].

III. THE REDUCED LWR

First, exploiting a ring homomorphism, we can assume that $q = p + 1$. In fact, given the ring homomorphism $\phi_{q,p+1} :$

¹Serial implementations are expected to dominate in practice, as parallel LWR designs incur significant hardware cost and most target platforms are resource-constrained.

REFERENCES

- [1] P. Kocher, J. Jaffe, and B. Jun, “Differential power analysis,” in *Annual international cryptology conference*, pp. 388–397, Springer, 1999.
- [2] S. Chari, C. S. Jutla, J. R. Rao, and P. Rohatgi, “Towards sound approaches to counteract power-analysis attacks,” in *Annual International Cryptology Conference*, pp. 398–412, Springer, 1999.
- [3] G. Cassiers, B. Grégoire, I. Levi, and F.-X. Standaert, “Hardware private circuits: From trivial composition to full verification,” *IEEE Transactions on Computers*, vol. 70, no. 10, pp. 1677–1690, 2020.
- [4] I. Levi and O. Keren, “Consolidated linear masking (clm): Generalized randomized isomorphic representations, powerful degrees of freedom and low (er)-cost,” *Cryptology ePrint Archive*, 2024.
- [5] F. Berti, S. Faust, and M. Orlt, “Provable secure parallel gadgets,” *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, vol. 2023, no. 4, pp. 420–459, 2023.
- [6] A. Banerjee, C. Peikert, and A. Rosen, “Pseudorandom functions and lattices,” in *EUROCRYPT*, 2012.
- [7] S. Dziembowski, S. Faust, G. Herold, A. Journault, D. Masny, and F. Standaert, “Towards sound fresh re-keying with hard (physical) learning problems,” in *CRYPTO*, 2016.
- [8] D. Kamel, F. Standaert, A. Duc, D. Flandre, and F. Berti, “Learning with physical noise or errors,” *IEEE Trans. Dependable Secur. Comput.*, vol. 17, no. 5, pp. 957–971, 2020.
- [9] D. Bellizia, N. E. Mrabet, A. P. Fournaris, S. Pontié, F. Regazzoni, F. Standaert, É. Tasso, and E. Valea, “Post-quantum cryptography: Challenges and opportunities for robust and secure HW design,” in *36th IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems, DFT 2021, Athens, Greece, October 6-8, 2021* (L. Dilillo, L. Cassano, and A. Papadimitriou, eds.), pp. 1–6, IEEE, 2021.
- [10] D. Bellizia, C. Hoffmann, D. Kamel, P. Méaux, and F. Standaert, “When bad news become good news towards usable instances of learning with physical errors,” *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, vol. 2022, no. 4, pp. 1–24, 2022.
- [11] D. Kamel, D. Bellizia, F. Standaert, D. Flandre, and D. Bol, “Demonstrating an LPPN processor,” in *Proceedings of the 2018 Workshop on Attacks and Solutions in Hardware Security, ASHES@CCS 2018, Toronto, ON, Canada, October 19, 2018* (C. Chang, U. Rührmair, D. E. Holcomb, and J. Guajardo, eds.), pp. 18–23, ACM, 2018.
- [12] D. Boneh, R. A. DeMillo, and R. J. Lipton, “On the importance of checking cryptographic protocols for faults (extended abstract),” in *Advances in Cryptology - EUROCRYPT '97, International Conference on the Theory and Application of Cryptographic Techniques, Konstanz, Germany, May 11-15, 1997, Proceeding* (W. Fumy, ed.), vol. 1233 of *Lecture Notes in Computer Science*, pp. 37–51, Springer, 1997.
- [13] S. P. Skorobogatov and R. J. Anderson, “Optical fault induction attacks,” in *Cryptographic Hardware and Embedded Systems - CHES 2002, 4th International Workshop, Redwood Shores, CA, USA, August 13-15, 2002, Revised Papers* (B. S. K. Jr., Ç. K. Koç, and C. Paar, eds.), vol. 2523 of *Lecture Notes in Computer Science*, pp. 2–12, Springer, 2002.
- [14] M. Dehbaoui, J. Dutertre, B. Robisson, and A. Tria, “Electromagnetic transient faults injection on a hardware and a software implementations of aes,” in *FDTC*, pp. 7–15, 2012.
- [15] S. Ordas, B. Zussa, and P. Maurine, “Electromagnetic fault injection: Towards a fault model on a 32-bit microcontroller,” in *FDTC*, pp. 77–86, 2014.
- [16] F. Berti and F. Standaert, “An analysis of the learning parity with noise assumption against fault attacks,” in *CARDIS*, vol. 10146 of *Lecture Notes in Computer Science*, pp. 245–264, Springer, 2016.